Introduction

We hope to solve the equation

$$i\partial_t \psi = (-i\sigma_x \partial_x + m\sigma_z + f(x))\psi$$

where $\psi = (\psi_1 \psi_2)^T$, and f is some potential function. In theory, this is only slightly different from Schrödinger. If we can store the spinor ψ, nearly all the same methods can be used to solve Dirac; we will use Strang splitting to approximate the solution by solving each term individually. Note that $\propto$ can usually be interpreted as "equal up to normalization."

Discretization and Storage

Fortunately, the problem of storing the spinor on qubits has a simple solution: we maintain a register to store position, as in Schrödinger, and add one qubit as an indicator. We discretize ψ_1 and ψ_2 into $|\psi_1\rangle$ and $|\psi_2\rangle$ as we know how from Schrödinger, and store the state ψ as

$$\begin{aligned}
|\psi\rangle &\propto \sum_{k=0}^{N-1} (\psi_1(x_k)\,|0\rangle + \psi_2(x_k)\,|1\rangle) \otimes |k\rangle \\
&= |0\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_1(x_k)\,|k\rangle\right) + |1\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_2(x_k)\,|k\rangle\right) \\
&\propto |0\rangle \otimes |\psi_1\rangle + |1\rangle \otimes |\psi_2\rangle
\end{aligned}$$

With the indicator as the most significant qubit, this is equivalent to concatenating the column vectors of amplitudes for $|\psi_1\rangle$ and $|\psi_2\rangle$. Regarding measurement probabilities, if we measure only the register storing position, and leave the indicator qubit unmeasured, the probability of measuring $|k\rangle$ is $p(k) \propto |\psi_1(x_k,t)|^2 + |\psi_2(x_k),t|^2$. This is justified as follows: the state $|\psi\rangle$ has density matrix

$$
\begin{aligned}
|\psi\rangle\langle\psi| &\propto (|0\rangle\otimes|\psi_1\rangle + |1\rangle\otimes|\psi_2\rangle)(\langle 0|\otimes\langle\psi_1| + \langle 1|\otimes\langle\psi_2|) \\
&= |0\rangle\langle 0|\otimes|\psi_1\rangle\langle\psi_1| + |0\rangle\langle 1|\otimes|\psi_1\rangle\langle\psi_2| + |1\rangle\langle 0|\otimes|\psi_2\rangle\langle\psi_1| + |1\rangle\langle 1|\otimes|\psi_2\rangle\langle\psi_2|
\end{aligned}
$$

Measurement probabilities of the position register are given by the reduced density matrix obtained by tracing out the spinor qubit. Taking the partial trace, we get

$$
\begin{aligned}
\rho &\propto |\psi_1\rangle\langle\psi_1| + |\psi_2\rangle\langle\psi_2| \\
&= \sum_{l=0}^{N-1}\sum_{j=0}^{N-1}(\psi_1(x_l)\psi_1(x_j)^* + \psi_2(x_l)\psi_2(x_j)^*)\,|l\rangle\langle j|
\end{aligned}
$$

Thus the probability of measuring $|k\rangle$ at time t is $p(k) = \mathrm{tr}(|k\rangle\langle k|\,\rho) \propto |\psi_1(x_k,t)|^2 + |\psi_2(x_k,t)|^2$ by orthonormality of the computational basis.

Transport Term

We concern ourselves with the equation $i\partial_t\psi = -i\partial_x\sigma_x\psi$. Recognizing that $\sigma_x = H\sigma_z H$, where, H is the Hadamard matrix, this system is equivalent to $\partial_t H\psi = -\partial_x\sigma_z H\psi$. Letting $\varphi := (\varphi_1,\varphi_2)^T = H\psi$, since σ_z is diagonal, the system is decoupled: $\partial_t\varphi = -\partial_x\sigma_z\varphi$. Equivalently,

$$
\begin{aligned}
\partial_t\varphi_1 &= -\partial_x\varphi_1 \\
\partial_t\varphi_2 &= \partial_x\varphi_2
\end{aligned}
$$

This is a first order linear transport equation, which has a known solution for initial conditions $\varphi_j^0(x) = \varphi_j(x,0)$: $\varphi_1(x,t) = \varphi_1^0(x-t)$ and $\varphi_2(x,t) = \varphi_2^0(x+t)$. For the sake of implementation on a quantum computer, we solve using the

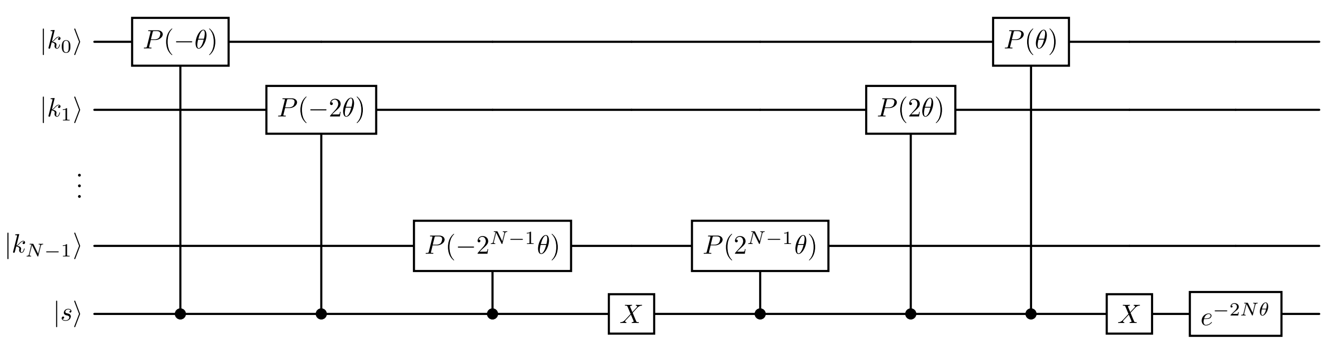
Fourier transform $\mathcal{F}$ to get

$$\varphi(x,t) = \begin{bmatrix} \mathcal{F}^{-1}(e^{i\xi t}\mathcal{F}(\varphi_1^0)(\xi)) \\ \mathcal{F}^{-1}(e^{-i\xi t}\mathcal{F}(\varphi_2^0)(\xi)) \end{bmatrix}$$

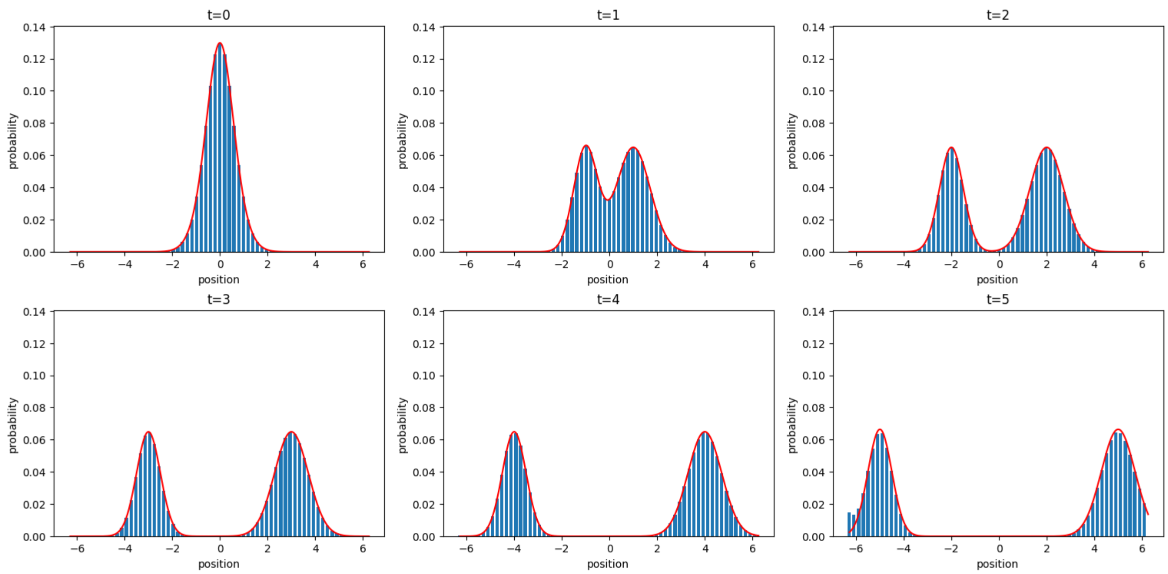
Hence on the quantum computer we map $|s\rangle \otimes |k\rangle \mapsto |s\rangle \otimes (QFT e^{(-1)^s i\xi_k \Delta t} QFT^\dagger |k\rangle)$, where as before, $\xi_k = \frac{\pi}{L}(2k - N)$. A new concern we have in Dirac is that any θ dependent global phase incurred in the application of $e^{i\theta k}$ is now a relative phase between φ_1 and φ_2 that we need to correct for. We write the phase shift operation more carefully with respect to global phase changes. Note that I've changed the implementation of $e^{ik} := \bigotimes_{j=0}^{n-1} P(2^j)$ to use phase gates, rather than differences of identity and Z gates. This makes global phase easier to keep track of.

$$\begin{aligned} \exp{(-1)^s \frac{\pi}{L}(2k-N)\Delta t} &= \exp\left((-1)^{s+1} iN\frac{\pi}{L}\Delta t\right) \exp\left((-1)^s 2i\frac{\pi}{L}\Delta t k\right) \\ &= \exp\left((-1)^{s+1} iN\frac{\pi}{L}\Delta t\right) \bigotimes_{j=0}^{n-1} P\left((-1)^s 2^{j+1}\frac{\pi}{L}\Delta t\right) \end{aligned}$$

This gives us a relative phase difference of $e^{\frac{2\pi i N\Delta t}{L}}$ between φ_1 and φ_2, which we compensate for by applying $P\left(\frac{-2\pi N\Delta t}{L}\right)$ to the spinor qubit. The quantum circuit we implement to apply the shift phase is below. $\theta = \frac{\pi}{L}\Delta t$.



Sandwiching this with QFT and $QFT^\dagger$ solves the linear transport equation, which is certainly interesting on its own, so I have included a plot below of a solution with two Gaussians φ_1 and φ_2.



Test